

Cyber Security - 100 Hours - Session Plan

Stage 1

Cybersecurity Threats, Vulnerabilities, and Attacks	Introduction	Lecture
	Common threats	Lecture
	Deception	Lecture
	Cyber Attacks	Lecture
	Wireless and Mobile Device Attacks	Lecture
	Application Attacks	Lecture
	Cybersecurity Threats, Vulnerabilities, and Attacks Summary	Lecture
Securing Networks	Introduction	Lecture
	Current State of Affairs	Lecture
	Who is attacking our Network	Lecture
	Securing Networks Summary	Lecture
Attacking the Foundation	Introduction	Lecture
	IP PDU Details	Lecture
	IP Vulnerabilities	Lecture
	TCP and UDP Vulnerabilities	Lecture
	Attacking the Foundation Summary	Lecture
Attacking What We Do	Introduction	Lecture
	IP Services	Lecture
	Enterprise Services	Lecture
	Mitigating Common Network Attacks	Lecture
	Attacking What We Do Summary	Lecture
Wireless Network Communication	Introduction	Lecture
	Wireless Communication	Lecture
	WLAN Threats	Lecture
	Secure WLANs	Lecture
	Wireless Network Communication Summary	Lecture
Network Security Infrastructure	Introduction	Lecture
	Security Devices	Lecture
	Security Services	Lecture
	Network Security Infrastructure Summary	Lecture
The Windows Operating System	Introduction	Lecture
	Windows History	Lecture
	Windows Architecture and Operations	Lecture
	Windows Configuration and Monitoring	Lecture
	Windows Security	Lecture
	The Windows Operating System Summary	Lecture

Linux Overview	Introduction	Lecture
	Linux Basics	Lecture
	Working in the Linux Shell	Lecture
	Linux Server and Clients	Lecture
	Basic Server Administration	Lecture
	The Linux File System	Lecture
	Working with the Linux GUI	Lecture
	Working on a Linux Host	Lecture
	Linux Basics Summary	Lecture
System and Endpoint Protection	Introduction	Lecture
	Defending System and Devices	Lecture
	Antimalware Protection	Lecture
	Host-Based Intrusion Prevention	Lecture
	Application Security	Lecture
	System and Endpoint Protection Summary	Lecture
Stage-1 Revision - Part 1		
Stage-1 Revision - Part 2		
Formative Assessment - 1		

Stage 2

Cybersecurity Principles, Practices, and Processes	Introduction	Lecture
	The Three Dimensions	Lecture
	States of Data	Lecture
	Cybersecurity Principles, Practices, and Processes Summary	Lecture
Understanding Defense	Introduction	Lecture
	Defense-in-Depth	Lecture
	Cybersecurity Operations Management	Lecture
	Security Policies, Regulations, and Standards	Lecture
	Understanding Defense Summary	Lecture
System and Network Defense	Introduction	Lecture
	Physical Security	Lecture
	Application Security	Lecture
	Network Hardening: Services and Protocols	Lecture
	Network Hardening: Segmentation	Lecture
	Hardening Wireless and Mobile Devices	Lecture
	Cybersecurity Resilience	Lecture
	Embedded and Specialized Systems	Lecture
	System and Network Defense Summary	Lecture
Access Control	Introduction	Lecture
	Access Controls	Lecture
	Access Controls Concepts	Lecture
	Account Management	Lecture
	AAA Usage and operations	Lecture
	Access Control Summary	Lecture

Access Control Lists	Introduction	Lecture
	Introduction to Access Control Lists	Lecture
	Wildcard Masking	Lecture
	Configure ACLs	Lecture
	Named Standard IPv4 ACL Syntax	Lecture
	Implement ACLs	Lecture
	Mitigate Attacks with ACLs	Lecture
	IPv6 ACLs	Lecture
	Access Control Lists Summary	Lecture
Firewall Technologies	Introduction	Lecture
	Secure Networks with Firewalls	Lecture
	Firewalls in Network Design	Lecture
	Firewall Technologies Summary	Lecture
Zone-Based Policy Firewalls	Introduction	Lecture
	ZPF Overview	Lecture
	ZPF Operation	Lecture
	Configure a ZPF	Lecture
	Zone-Based Policy Firewalls Summary	Lecture
Cloud Security	Introduction	Lecture
	Virtualisation and Cloud Computing	Lecture
	The Domains of Cloud Security	Lecture
	Cloud Infrastructure Security	Lecture
	Cloud Application Security	Lecture
	Cloud Data Security	Lecture
	Protecting VMs	Lecture
	Cloud Security Summary	Lecture
Stage-2 Revision - Part 1		
Stage-2 Revision - Part 2		
Formative Assessment - 2		

Stage 3

Cryptography	Introduction	Lecture
	Confidentiality	Lecture
	Obscuring Data	Lecture
	Integrity and Authenticity	Lecture
	Using Hashes	Lecture
	Public Key Cryptography	Lecture
	Authorities and the PKI Trust System	Lecture
	Applications and Impacts of Cryptography	Lecture
	Cryptography Summary	Lecture
Technologies and Protocols	Introduction	Lecture
	Monitoring Common Protocols	Lecture
	Security Technologies	Lecture
	Technologies and Protocols Summary	Lecture

Network Security Data	Introduction	Lecture
	Types of Security Data	Lecture
	End Device Logs	Lecture
	Network Logs	Lecture
	Network Security Data Summary	Lecture
Evaluating Alerts	Introduction	Lecture
	Sources of Alerts	Lecture
	Overview of Alert Evaluation	Lecture
	Evaluating Alerts Summary	Lecture
Governance and Compliance	Introduction	Lecture
	Governance	Lecture
	The Ethics of Cybersecurity	Lecture
	IT Security Management Framework	Lecture
	Governance and Compliance Summary	Lecture
Network Security Testing	Introduction	Lecture
	Security Assessments	Lecture
	Network Security Testing Techniques	Lecture
	Network Security Testing Tools	Lecture
	Penetration Testing	Lecture
	Network Security Testing Summary	Lecture
Threat Intelligence	Introduction	Lecture
	Information Sources	Lecture
	Threat Intelligence Services	Lecture
	Threat Intelligence Summary	Lecture
Endpoint Vulnerability Assessment	Introduction	Lecture
	Network and Server Profiling	Lecture
	Common Vulnerability Scoring System (CVSS)	Lecture
	Secure Device Management	Lecture
	Endpoint Vulnerability Assessment Summary	Lecture
Risk Management and Security Controls	Introduction	Lecture
	Risk Management	Lecture
	Risk Assessment	Lecture
	Security Controls	Lecture
	Risk Management and Security Controls Summary	Lecture
Digital Forensics and Incident Analysis and Response	Introduction	Lecture
	Evidence Handling and Attack Attribution	Lecture
	The Cyber Kill Chain	Lecture
	The Diamond Model of Intrusion Analysis	Lecture
	Incident Response	Lecture
	Disaster Recovery	Lecture
	Digital Forensics and Incident Analysis and Response Summary	Lecture
Stage-3 Revision - Part 1		
Stage-3 Revision - Part 2		
Formative Assessment - 3		

Stage 4

Governance and Compliance	Introduction	Lecture
	Governance	Lecture
	The Ethics of Cybersecurity	Lecture
	IT Security Management Framework	Lecture
	Governance and Compliance Summary	Lecture
Network Security Testing	Introduction	Lecture
	Security Assessments	Lecture
	Network Security Testing Techniques	Lecture
	Network Security Testing Tools	Lecture
	Penetration Testing	Lecture
	Network Security Testing Summary	Lecture
Threat Intelligence	Introduction	Lecture
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	Threat Intelligence Services	Lecture
	Threat Intelligence Summary	Lecture
Endpoint Vulnerability Assessment	Introduction	Lecture
	Network and Server Profiling	Lecture
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	Secure Device Management	Lecture
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Risk Management and Security Controls	Introduction	Lecture
	Risk Management	Lecture
	Risk Assessment	Lecture
	Security Controls	Lecture
	Risk Management and Security Controls Summary	Lecture
Digital Forensics and Incident Analysis and Response	Introduction	Lecture
	Evidence Handling and Attack Attribution	Lecture
	The Cyber Kill Chain	Lecture
	The Diamond Model of Intrusion Analysis	Lecture
	Incident Response	Lecture
	Disaster Recovery	Lecture
	Digital Forensics and Incident Analysis and Response Summary	Lecture
Full course Revision - Part 1		
Full course Revision - Part 2		
Full course Revision - Part 3		
Full course Revision - Part 4		
Full course Revision - Part 5		
Summative Assessment		